

Yingrui Hou

Ingénieure simulation & validation logicielle pour systèmes physiques

Target Application: Framatome | Ingénieur Python - Validation et Développement F/H

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Introduction personnelle

Ingénieure simulation et validation issue de la physique expérimentale, avec plusieurs années d'expérience dans le développement de workflows réutilisables en Python/C++ pour des systèmes de mesure et de simulation.

Je conçois des chaînes de validation, de calibration et de correction qui rendent des logiciels et des modèles plus fiables avant décision.

Mon point fort est de transformer des écarts entre simulation, mesure et logiciel en actions concrètes: validation, diagnostic, correction et amélioration continue en environnement technique partagé.

Selected Fit For This Role

- Developed reusable Python/C++ workflows for validation, calibration, and analysis of noisy measurement systems in shared technical environments.
- Built monitoring and validation workflows to identify reconstruction and calibration discrepancies, then trace anomalies back to detector or software causes.
- Identified concrete anomalies that led to corrective actions, including a missing-energy software issue later fixed in reconstruction.
- Developed ROOT/C++ calibration and correction workflows to reduce structured bias and compare performance before and after correction across operating conditions.
- Worked in large shared codebases with review discipline, result reproducibility, and reinjection of validation outputs into modelling decisions.
- Aligned simulation and data through reweighting and control-sample calibration before downstream decisions, avoiding the propagation of software or model bias.

software validation, automation, Python, Linux, critical environment, and a direct link to software quality and internal simulation tools.

Education

PhD in Particle Physics
University of Chinese Academy of Sciences
BSc in Applied Physics
China University of Mining and Technology

Languages

Chinese: Native speaker
English: C1
French: B1 to B2

Core Tools & Methods

Tools: ROOT / RooFit | Python | Geant4 | Workflow Automation (YAML / Shell / CMake) | NumPy | C/C++ | Geant4 simulation | pandas | statsmodels | matplotlib / seaborn | etc

Methods: Detector Simulation | response-map reconstruction | correlation-based correction | event-level correction | production yield workflows | Latent-Parameter Estimation | Maximum Likelihood Estimation | beam-test analysis | particle-dependent timing models | Probabilistic Modelling | etc

Validation: Calibration & Bias Correction | Model Validation | Simulation-to-Data Alignment | Simulation Calibration | Simulation Correction | Monitoring | Data Validation | Root-Cause Analysis | Bias Correction | Calibration | etc

ML: Feature Validation | train-test diagnostics | scikit-learn | CatBoost / Gradient Boosting | Domain Adaptation | GBDT | CatBoost | Reweighting | Feature Selection | CatBoost training | etc

Relevant Experience

Postdoctoral Researcher | 2021 - 2025

Inference, Validation & Analysis Workflows

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Built reusable C++/Python/ROOT workflows for weak-signal extraction, time-dependent inference, and validation on noisy datasets where direct event-level truth was unavailable.
- Implemented likelihood-based models and joint fits to estimate latent parameters under sample mismatch, finite resolution, and biased observation conditions.
- Aligned simulated and observed datasets through gradient-boosting reweighting and control-sample calibration before downstream inference and selection decisions.

Simulation, Calibration & Test-Data Interpretation

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Built time-resolved signal-analysis workflows that recovered component fractions from scintillation timing and improved reconstructed-energy resolution by about a factor of 2 in representative PSD studies.
- Developed ROOT/C++ calibration and correction workflows to reduce structured bias and compare performance before and after correction across operating conditions.
- Built benchmark-based monitoring workflows to validate reconstruction and calibration quality, then traced anomalies to detector behaviour or software issues instead of stopping at descriptive analysis.

Selected Project

Test-Data Validation & Root-Cause Analysis

Monitoring | Data Validation | Performance Diagnostics | Root-Cause Analysis

Built a production-style validation workflow for early Run 3 photon reconstruction, focused on finding actionable detector and software issues.

Methods: benchmark-channel checks, MC-data comparison, detector-level diagnostics

Impact: translated validation discrepancies into concrete calibration, detector, and software follow-ups before they became accepted bias.

Industry relevance: Production Monitoring | Model Validation | Observability | Quality Assurance

Sensor Simulation & Reconstruction

Signal Processing | Simulation Calibration | Digital Twins

Built a simulation-and-measurement workflow to quantify how detector non-uniformity affects energy resolution, using beam-test data to constrain the model instead of assuming ideal sensor...

Methods: Geant4 simulation, beam-test analysis, response-map reconstruction

Impact: linked measured non-uniformity to system-level resolution through both simulation code and experimental validation.

Industry relevance: Sensor Analytics | Measurement Systems | Simulation-Assisted Design

Pulse-Shape Discrimination for Energy Reconstruction

Pulse Shape Discrimination | Detector Simulation | Time-Resolved Signal Analysis | Resolution Optimisation

Built a PSD-oriented simulation and analysis workflow to extract shower composition from scintillation time profiles and use it to improve reconstructed energy.

Methods: particle-dependent timing models, component fitting, correlation-based correction

Impact: showed that PSD-derived shower fractions can improve reconstructed energy and help judge which detector materials remain usable for timing-based...

Industry relevance: Time-Resolved Sensing | Signal Decomposition | Measurement-System Design